

E-Waste & Environmental Management

Activity Worksheet

The Hidden Carbon Footprint of Our Gadgets

Name: Pranitha ShettyRoll No.: 24101B0027Division: BDate: 5/8/2026

Part A: Carbon Footprint Calculation

Use the carbon footprint calculator provided by your faculty.

Parameter	Your Response
Annual Carbon Footprint	<u>2.8</u> tCO ₂ e / year
National Average	<u>1.9</u> tCO ₂ e / year
Global Average	<u>4.7</u> tCO ₂ e / year
Number of Earths Required (if provided)	<u>1.7 Earths</u>

Which category contributed the most?

☐ Transportation ☐ Home Energy ☐ Food ☐ Shopping & Consumption

☒ Electronics / Digital Lifestyle ☐ Other: _____

Which result surprised you the most?

I was surprised that using electronic devices and internet services contributes significantly to my carbon footprint.

Part B: My Electronic Lifestyle

1. How many smartphones have you owned in the last 10 years?

☐ 1 ☒ 2-3 ☐ 4-5 ☐ More than 5

2. How often do you replace your smartphone?

☐ Every year ☐ Every 2 years ☒ Every 3-4 years ☐ More than 5 years

3. What do you usually do with your old phone?

☐ Keep ☐ Sell ☒ Exchange ☐ Donate

☐ Recycle ☐ Throw away

4. Have you ever repaired an electronic device instead of replacing it?

☒ Yes ☐ NoReplaced the charging cable / battery or repaired the screen
If yes: of my smartphone

5. Have you ever recycled e-waste through an authorized recycler?

☐ Yes ☒ No

Item: _____

Unused Electronic Items at Home	Number
Mobile phones	<u>1</u>
Chargers	<u>2</u>
Earphones	<u>2</u>
Batteries	<u>2</u>
Cables	<u>3</u>
Other	<u>-</u>

Part C: Think Like an Environmental Engineer

Habit	Higher Carbon Footprint	More E-Waste
Buying a new phone every 2 years	☒	☒
Repairing a laptop	☐	☐
Keeping old phones unused in drawers	☐	☒
Recycling electronic devices	☐	☐
Using devices for 5-6 years	☐	☐

Which stage of an electronic product's life cycle contributes the MOST carbon emissions?

☐ Raw material extraction ☒ Manufacturing ☐ Transportation☐ Usage ☐ Disposal

Why is extending the life of an electronic device environmentally beneficial?

Extending the life of electronic devices reduces e-waste
conserves natural resources, lowers manufacturing related
carbon emissions.

Part D: Reflection

Three habits that increase my personal carbon footprint:

- ① charging electronic devices frequently.
- ② Using electronic devices for long hours.
- ③ Buying new gadgets before the old ones wear out.

Three habits that increase e-waste generation:

- ① Replacing devices too often
- ② Keeping old electronics unused instead of recycling them.
- ③ Throwing electronic accessories in regular waste

Action to Reduce Carbon Emissions & E-waste	Expected Impact
Use devices for 5-6 years before replacing them.	Reduces carbon emissions and e-waste
Repair devices instead of buying new ones.	Extends product life and conserves resources.
Recycle e-waste through authorized recyclers.	Recovers valuable materials and reduces pollution.

Discussion Questions (Optional)

1. Which stage of an electronic product's life cycle has the highest carbon footprint and why?

The manufacturing stage has the highest carbon footprint because extracting raw materials and producing electronic components.

2. How does extending the lifespan of smartphones reduce both carbon emissions and e-waste?

Using smartphones for a longer time reduces the need to manufacture new devices, which lowers carbon emission and decrease the amt of e-waste.

3. Why is recycling alone not enough for sustainable e-waste management?

Recycling cannot recover all materials and still requires energy. Reducing consumption, reusing and repairing devices.

4. Explain how the 5Rs contribute to lowering the environmental impact of electronic products.

Refuse, Reduce, Reuse, Repair and Recycle - help conserve natural resources, reduce carbon emissions.

5. As an engineering student, what innovative solution would you propose to reduce the carbon footprint associated with electronic devices?

I would develop an AI based app that reminds users to repairs and maintain devices, tracks battery health and suggests nearby e-waste recycling centers.

This would increase device lifespan & reduce carbon emission.

Ms. Nilima Main

Assistant Professor, Department of First Year Engineering
Vidyalankar Institute of Technology